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Anastasiya_Proposal

Assignment 202

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IARCO RESEARCH PROPOSAL

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Research Topic: The Role of Reflective Prompts in Evaluating Social Media Algorithms' Impact on Adolescent Attention Span: A Controlled Intervention Study

Title: The Role of Reflective Prompts in Evaluating Social Media Algorithms' Impact on Adolescent Attention Span: A Controlled Intervention Study

Research Problem

In recent years, social media has established itself as a primary entertainment source globally, with usage accelerating during the COVID-19 pandemic. While offering connectivity, its impact on cognitive regulation presents critical challenges. Li et al. [1] describe reduced metacognitive monitoring accuracy among adolescents engaged in media multitasking, leaving young users vulnerable to inattention driven by algorithmically curated content. While platform-induced inattention is widely documented, limited empirical research examines whether structured metacognitive interventions can alter how teenagers perceive and regulate their attention within digital environments. This study addresses this gap by assessing whether a controlled reflective intervention can change adolescent perception and behavioral self-regulation of attention.

Literature Review

Digital platform engagement correlates strongly with adolescent cognitive distraction, where problematic usage patterns — rather than mere exposure duration — exacerbate ADHD symptoms [2], [3]. Short-form video structures amplify addictive behaviors; Chen et al. [4] observed that addicted users exhibit pronounced attention deficits during short-form video consumption. These platforms use AI algorithms designed to maximize engagement by personalizing feeds, often without ethical considerations, which reinforces behavioral addiction [4].

To mitigate digital overuse, traditional interventions have focused on mindfulness and total restriction. Mindfulness programs, such as Mindful Connections, demonstrate efficacy in reducing problematic usage and improving emotional regulation among adolescents [5], [6]. Conversely, broad screen-time reduction initiatives frequently yield negligible long-term results, with total digital detoxes proving socially disruptive and unsustainable [7]. While structured reflection is validated in educational settings for enhancing focus and linguistic metacognition [8], its application toward evaluating personalized algorithms remains unexplored.

Research Questions and Hypotheses

This study investigates whether daily reflective prompts influence adolescents' awareness of algorithmic content curation and whether this awareness corresponds to measurable shifts in attention regulation.

Research Questions:

1. How do daily reflective prompts affect adolescent awareness regarding:

a. Emotional and somatic responses during platform disconnection?
b. Algorithmic retention design (e.g., recommendation loops, infinite scroll)?
c. Metacognitive evaluation of sustained focus?

2. Is there a statistically significant difference between the experimental and control groups regarding:

a. Perceived sustained focus during non-digital tasks?

b. Daily social media screen time duration?

3. Is there a significant correlation between increases in qualitative reflection scores and changes in daily social media screen time?

Hypotheses:

· H1: Exposure to reflective prompts will yield a statistically significant increase in adolescent algorithmic awareness compared to a control condition.
· H2: Increased algorithmic awareness will correspond to measurable improvements in self-reported attention regulation and intentionality of device usage.

Proposed Methodology

This study will employ a 7-day convergent mixed-methods randomized controlled design. To isolate the cognitive effect of reflection from physical phone separation, both experimental and control conditions will undergo an identical daily off-screen challenge.

1. Participants and Sampling

Fifty adolescents (aged 13–17) will be recruited via targeted social media sampling. Inclusion criteria require daily social media usage and access to a smartphone with built-in screen time tracking. Following parental consent and participant assent, subjects will be randomly assigned to the Experimental Group (n = 25) or the Control Group (n = 25).

2. Intervention Procedures

Both groups will complete a daily 15–20 minute off-screen activity without their phones. Data submission will occur immediately post-activity via secure digital forms:

Experimental Condition: Participants will execute the daily off-screen challenge and respond to a structured metacognitive prompt protocol:

· Day 1 (Emotions): "What did you notice about your emotions during the phone-free time?"
· Day 2 (Physical Sensations): "What did you feel in your body during this time?"
· Day 3 (Thoughts & Distractions): "What thoughts or distractions came up? Did any social media content seem designed to keep you scrolling, and did it affect how long you could stay focused?"
· Day 4 (Successes & Difficulties): "What was easy? What was difficult? How long were you able to stay focused without checking your phone?"
· Day 5 (Comparison): "How does your ability to concentrate today compare to the beginning of the week?"
· Day 6 (Patterns & Algorithms): "Do you notice any repeating patterns in your distractions? Do you think social media algorithms influenced how long you were able to concentrate today?"
· Day 7 (Overall Reflection): "What has changed in how you perceive your attention span? Has your awareness of how algorithms affect your attention changed?"

Control Condition: Participants complete the identical off-screen activity but submit a neutral activity log (e.g., "What activity did you do during your phone-free time today?") to isolate the physical effect of phone separation.

3. Measures and Operationalization

Operationalization of Attention Span: Perceived sustained focus — specifically, an adolescent's self-reported capacity to remain engaged in a non-digital activity without checking their phone — supplemented by behavioral app usage metrics.

Screen Time Data (Quantitative): Daily system-generated screenshots (Apple Screen Time or Android Digital Wellbeing) tracking time spent specifically in "Social" applications (e.g., TikTok, Instagram, YouTube Shorts), captured pre-challenge and the following morning.

Reflective Data (Qualitative): Daily written reflections or activity logs submitted alongside screenshots via secure forms.

4. Data Integration and Analysis Plan

Qualitative Scoring (Quantitizing): Reflections will undergo thematic analysis to construct a 5-point ordinal scale (1 = Low Algorithmic Awareness, 5 = High Metacognitive Evaluation). Inter-rater reliability will be evaluated across dual-coded sub-samples.

Quantitative Comparison: Changes in daily social media screen time (Δ Screen Time) over 7 days will be evaluated using Repeated-Measures ANOVA (or the non-parametric Friedman Test if normality is violated).

Mixed-Methods Correlation: Spearman's Rank Correlation (ρ) will test the association between changes in reflection scores and changes in daily social media usage within the experimental group.

Expected Outcomes

The intervention is expected to demonstrate that reflective prompts elevate adolescent awareness regarding algorithmic influence on focus and mood. Increased metacognitive awareness is expected to foster greater intentionality in screen engagement rather than a drastic overall reduction in screen time. Differences between experimental and control groups are anticipated to be moderate, reflecting deeper cognitive processing in the experimental condition.

Potential Limitations

· Sample Constraints: A sample size of N = 50 and a 7-day duration limit broad generalizability and long-term behavioral assessment.
· Self-Report Bias: Reliance on subjective responses introduces potential social desirability bias and reporting inaccuracies.
· External Confounders: Variations in baseline digital literacy, academic stress, or family context may influence attention capacity.
· Methodological Controls: Measurement focuses on perceived attention rather than laboratory-based neurocognitive metrics. Qualitative rating subjectivity will be managed using pre-defined rubrics and inter-rater verification.

Ethical Considerations

The protocol adheres to ethical standards for minor participants. Electronic parental consent and participant assent will be gathered prior to participation. Data will be anonymized using pseudonyms, and app usage screenshots will be processed strictly for data extraction and kept confidential. Participation will be fully voluntary with unpenalized withdrawal options. Participants completing the 7-day protocol will receive a certificate of completion; no financial compensation will be provided.

Project Practicalities

The research project will be executed over a 4-month timeline:

· Months 1–2: Finalization of intervention materials, ethical approval acquisition, recruitment setup, and consent collection.
· Month 3: Execution of the 7-day intervention, daily data collection, and screenshot validation.
· Month 4: Qualitative thematic coding, statistical data analysis (ANOVA, Spearman ρ), synthesis of findings, and manuscript preparation.

Post-Program Plan

Upon completion, the study findings will be prepared for submission to academic conferences and peer-reviewed journals specializing in educational technology or behavioral psychology. The findings will also be leveraged to assist secondary schools in developing evidence-based digital literacy and metacognitive reflection modules for teenage students.

References

[1] S. Li, Y. Liu, A. Jing, and Y. Wang, "The effects of multitasking on metacognitive monitoring in primary and secondary school students," Journal of Experimental Child Psychology, vol. 242, p. 105908, Jun. 2024. DOI: 10.1016/j.jecp.2024.105908

[2] T. Siebers, I. Beyens, J. L. Pouwels, and P. M. Valkenburg, "Social media and distraction: An experience sampling study among adolescents," Media Psychology, vol. 25, no. 3, pp. 343–366, 2022. DOI: 10.1080/15213269.2021.1959350

[3] M. Boer, G. Stevens, C. Finkenauer, and R. van den Eijnden, "Attention deficit hyperactivity disorder-symptoms, social media use intensity, and social media use problems in adolescents: Investigating directionality," Child Development, vol. 91, no. 4, pp. e853–e865, Jul./Aug. 2020. DOI: 10.1111/cdev.13334

[4] Y. Chen, M. Li, F. Guo, and X. Wang, "The effect of short-form video addiction on users' attention," Behaviour & Information Technology, vol. 42, no. 16, pp. 2893–2910, 2023. DOI: 10.1080/0144929X.2022.2151512

[5] J. L. Weaver and J. M. Swank, "Mindful Connections: A mindfulness-based intervention for adolescent social media users," Journal of Child and Adolescent Counseling, vol. 5, no. 2, pp. 103–112, 2019. DOI: 10.1080/23727810.2019.1586419

[6] S. Ünlü, K. Uzun, and G. Arslan, "Mindfulness-based intervention in schools: Addressing social media burnout and enhancing well-being in adolescents," Children, vol. 12, no. 7, p. 826, 2025. DOI: 10.3390/children12070826

[7] E. C. Sullivan, S. A. Cairney, and L.-M. Henderson, "Innovations in Practice: A pilot randomised trial on smartphone and social media abstinence – effects on sleep quality and psychological wellbeing in adolescents," Child and Adolescent Mental Health, 2026, DOI: 10.1111/camh.70118.

[8] P. Dao, M. X. N. C. Nguyen, and D.-N. Chi, "Reflective learning practice for promoting adolescent EFL learners' attention to form," Innovation in Language Learning and Teaching, vol. 15, no. 3, pp. 247–262, 2021. DOI: 10.1080/17501229.2020.1766467